

Topic 6: Chemical reactions

Premium Answer Booklet

18 questions	2020-2025 Paper 4	Premium readable layout
-----------------	----------------------	----------------------------

Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

Contents

No	Question source	Marks
01	2020 May/June Paper 42 Question 4	16
02	2021 Feb/March Paper 42 Question 3	12
03	2021 May/June Paper 41 Question 4	7
04	2021 May/June Paper 43 Question 4	7
05	2021 Oct/Nov Paper 43 Question 6	10
06	2022 Feb/March Paper 42 Question 2	14
07	2022 Oct/Nov Paper 41 Question 3	14
08	2022 Oct/Nov Paper 43 Question 4	12
09	2023 May/June Paper 41 Question 4	14
10	2023 May/June Paper 42 Question 3	17
11	2023 May/June Paper 43 Question 4	10
12	2023 Oct/Nov Paper 41 Question 5	12
13	2024 May/June Paper 41 Question 4	10
14	2024 May/June Paper 43 Question 4	10
15	2024 Oct/Nov Paper 42 Question 3	19
16	2024 Oct/Nov Paper 43 Question 5	15
17	2025 Feb/March Paper 42 Question 4	14
18	2025 Oct/Nov Paper 43 Question 4	17

Question 01

16 marks

2020 May/June Paper 42 Question 4

Main topic: Topic 6: Chemical reactions
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s20_ms_42.pdf

4(a)	• substance that speeds up a reaction / increases rate (1) • unchanged (chemically) at the end OR not used up OR lowers activation energy OR provides alternative pathway (1)
4(b)	• rate decreases (1) • particles further apart / less particles per unit volume (1) • fewer collisions per unit time / lower collision frequency (1) • reaction stops because all hydrogen peroxide is used up
4(c)	• steeper gradient (1) • reaches same volume of oxygen (1)
4(d)	• particles gain kinetic energy / particles move faster (1) • greater number of collisions with activation energy (or more) / greater number of particles with activation energy (or more) / greater number of particles with energy required for reaction (1) • more collisions are successful / more collisions are fruitful / more collisions lead to reaction (1)
4(e)	• moles of oxygen • 48.0 • = • or moles of oxygen = 0.002 (1) • moles of hydrogen peroxide = x 2 or moles of hydrogen peroxide = 0.004 (1) • concentration = x 40 = 0.16 mol / dm³ (1) • allow ECF
4(f)	• $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ • ALL formulae correct (1) • balancing (1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 02

12 marks

2021 Feb/March Paper 42 Question 3

Main topic: Topic 6: Chemical reactions
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_m21_ms_42.pdf

3(a)(i)	• Haber
3(a)(ii)	• reversible / equilibrium
3(a)(iii)	• 450 °C (1) • 200 atm (1)
3(a)(iv)	• iron / Fe
3(a)(v)	• equilibrium shifts / moves in forwards direction (1) • fewer molecules / moles (of gas) on RHS (1)
3(a)(vi)	• particles have more energy (1) • more collisions (between particles) occur per second / per unit time (1) • a greater percentage / proportion / fraction of collisions (of particles) are successful / have energy above activation • energy / have energy equal to activation energy (1)
3(b)	• $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$ • $(\text{NH}_4)_2\text{SO}_4$ (1) correct equation (1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 03

7 marks

2021 May/June Paper 41 Question 4

Main topic: Topic 6: Chemical reactions
Answer source label: 0620_s21_ms_41.pdf

4(a)	- the rate of forward reaction equals the rate of the reverse reaction (1) - concentrations of reactants and products are constant (1)
4(b)(i)	(increased pressure) nitrogen dioxide particles or molecules (forced) closer together - OR - same number of nitrogen dioxide particles or molecules in a smaller volume
4(b)(ii)	fewer number of gas moles or molecules on left hand side or reactant side (of the equation) opposite reasoning also allowed
4(c)(i)	shifts to the right
4(c)(ii)	- increase / faster (1) - increase / faster (1)

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 04

7 marks

2021 May/June Paper 43 Question 4

Main topic: Topic 6: Chemical reactions
Answer source label: 0620_s21_ms_43.pdf

4(a)	- the rate of forward reaction equals (the rate of the) reverse reaction (1) - concentrations of reactants and products are constant (1)
4(b)(i)	same number of gas moles on both sides of the equilibrium / same number of gas molecules on both sides of the equilibrium
4(b)(ii)	iodine particles or molecules (forced) closer together / same number of iodine particles or molecules in a smaller volume
4(c)(i)	shifted to the left
4(c)(ii)	- decrease / slower (1) - decrease / slower (1) - Question Answer

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.

END OF ANSWER

Question 05

10 marks

2021 Oct/Nov Paper 43 Question 6

Main topic: Topic 6: Chemical reactions
Answer source label: 0620_w21_ms_43.pdf

6(a)	• (a substance that) speeds up a reaction / increases the rate of a reaction (1) • unchanged chemically at the end (1)
6(b)(i)	• oxygen gas escapes
6(b)(ii)	• rate decreases • OR • rate / reaction is fastest at the start (1) • reaction stops or • rate = zero (1)
6(c)(i)	• (particles) have more kinetic energy / particles move faster(1) • more collisions per unit time (1) • more particles have energy greater than or equal to activation energy • OR • more particles have sufficient energy to react • OR • greater percentage or greater proportion or greater fraction of collisions have sufficient energy to react • OR • greater percentage or greater proportion or greater fraction of collisions have energy greater than or equal to activation energy • OR • greater percentage or greater proportion or greater fraction of collisions are successful (1)
6(c)(ii)	• steeper gradient from start (1) • levels off at the same mass (1)

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 06

14 marks

2022 Feb/March Paper 42 Question 2

Main topic: Topic 6: Chemical reactions

Cross-topic links: Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer source label: 0620_m22_ms_42.pdf

2(a)	• $\text{MgCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$
2(b)	• eight crosses in second shell of Mg • 7 dots and 1 cross in third shell of both Cl • 2+ charge on Mg and - charge on both Cl ions on correct answer line
2(c)	• C atom double bonded to 2 O atoms • 4 non-bonding e- on each O and no non-bonding e- on C and both octets complete
2(d)(i)	• gradient (of line) decreases
2(d)(ii)	• concentration of particles (of acid) decreases • lower rate of collisions of particles • February/March 2022
2(d)(iii)	• a new line steeper than printed line and starts at origin and levels off earlier than printed line • levels off at the same volume
2(e)	• $\text{mol HCl} = 0.400 \times 25.0 / 1000 = 0.01(00)$ • $\text{mol CO}_2 = / 2 = 0.0100 / 2 = 0.005(00)$ • $\text{volume CO}_2 = \times 24000 = 120 \text{ (cm}^3\text{)}$

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 07

14 marks

2022 Oct/Nov Paper 41 Question 3

Main topic: Topic 6: Chemical reactions

Cross-topic links: Topic 5: Chemical energetics, Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer source label: 0620_w22_ms_41.pdf

3(a)	• Haber (process)
3(b)	• air
3(c)	• reversible
3(d)	• 450 (1) • 200 (1)
3(e)	• Iron / Fe
3(f)	• reduced temperature: • (position of) equilibrium moves to right-hand side (1) • reaction is exothermic (1) • reduced pressure: • (position of) equilibrium moves to left-hand side (1) • more (gaseous) moles on left hand side (1)
3(g)	• rate decreases and particles have less energy (1) • less collisions (between particles) occur per second / per unit time (1) • less of the particles/collisions have energy equal to or above the activation energy (1) • or • less of the particles / collisions have sufficient energy to react • or • a lower percentage / proportion / fraction of collisions (of particles) • are successful or • have energy equal to or above activation energy
3(h)	• (NH₄)₂SO₄

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 08

12 marks

2022 Oct/Nov Paper 43 Question 4

Main topic: Topic 6: Chemical reactions

Cross-topic links: Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer source label: 0620_w22_ms_43.pdf

4(a)(i)	• increases (1) • decreases (1) • increases (1) • increases (1)
4(a)(ii)	• forward reaction is endothermic
4(b)	• 3 dot and cross bonding pairs (1) • All lone pairs correct. • All octets should be complete AND dots and crosses should be different on P and all Cl (1)
4(c)	• $\text{P4O10} + 6\text{PCl5} \rightarrow 10\text{POCl3}$ • all formula correct and on the correct sides of the arrow (1) • equation fully correct (1)
4(d)	• H 3.66 / 1 • P 37.80 / 31 • O 58.54 / 16 • OR $\text{H} = 3.66$ $\text{P} = 1.22$ $\text{O} = 3.66$ • OR H:P:O 3:1:3 (1) • H3PO3 (1)
4(e)	• H3PO4 (1)

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.

END OF ANSWER

Question 09

14 marks

2023 May/June Paper 41 Question 4

Main topic: Topic 6: Chemical reactions
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s23_ms_41.pdf

4(a)	• increases the rate of reaction / • speeds up a reaction(1) • unchanged at the end of the reaction(1)
4(b)(i)	• oxygen escapes from the flask or apparatus
4(b)(ii)	• concentration of hydrogen peroxide is highest at the start / particles of hydrogen peroxide are closest together at the start • OR • collision frequency is highest at the start
4(b)(iii)	• the hydrogen peroxide is used up / • ALL the hydrogen peroxide has reacted or decomposed
4(c)	• kinetic energy of particles increases(1) • frequency of collisions between particles increases(1) • more or higher percentage or higher proportion or • higher fraction of particles have energy greater than / equal to activation energy • OR • more of the collisions or higher percentage or higher fraction of collisions have energy greater than or equal to activation • energy(1)
4(d)	• $(50.0 \rightarrow 0.200 / 1000 =) 0.01(1)$ • $0.005(1)$ • $0.16(0)(1)$
4(e)	• no effect
4(f)	• $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$ • all formulae correct(1) • equation correct(1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 10

17 marks

2023 May/June Paper 42 Question 3

Main topic: Topic 6: Chemical reactions

Cross-topic links: Topic 10: Chemistry of the environment, Topic 7: Acids, bases and salts

Answer source label: 0620_s23_ms_42.pdf

3(a)	• contact (process) • 3(b(i)) • burning sulfur (in air) • Roasting sulfide ores (in air)
3(b)(ii)	• the rate of forward reaction equals (the rate of the) reverse reaction • concentrations of reactants and products are constant
3(b)(iii)	• 450 (oC) • 200 (kPa) • vanadium(V) oxide
3(b)(iv)	• increases • increases • increase • no change
3(b)(v)	• kinetic energy of particles decreases • frequency of collisions between particles decreases • lower percentage / proportion / fraction of collisions / particles have energy greater than / qual to activation energy • OR • fewer of the collisions / particles have energy greater than / equal to activation energy
3(c)	• $4 \rightarrow \blacksquare 2$ or $\blacksquare 8$ • $S + (4 \rightarrow -2) = -2 \therefore S = + 6$

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 11

10 marks

2023 May/June Paper 43 Question 4

Main topic: Topic 6: Chemical reactions
Cross-topic links: Topic 3: Stoichiometry, Topic 7: Acids, bases and salts
Answer source label: 0620_s23_ms_43.pdf

4(a)(i)	• concentration (of sulfuric acid particles) decreases • OR • frequency of collisions between particles decreases
4(a)(ii)	• all the (sulfuric) acid has reacted
4(b)	• greater surface area (of zinc) • frequency of collisions between (zinc and acid) particles increases
4(c)	• $\text{mol H}_2\text{SO}_4 = 2.00 \times 25.0/1000 = 0.05(00)$ • $\text{mol H}_2 = 0.05(00)$ • $\text{vol H}_2 = 24 = 1.2(0)$
4(d)(i)	• $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$ • ZnCl_2 • equation correct
4(d)(ii)	• lighted splint and (squeaky) pop

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 12

12 marks

2023 Oct/Nov Paper 41 Question 5

Main topic: Topic 6: Chemical reactions
Cross-topic links: Topic 10: Chemistry of the environment
Answer source label: 0620_w23_ms_41.pdf

5(a)	- the rate of forward reaction equals (the rate of the) reverse reaction (1) - the concentrations of reactants and products are no longer changing (1)
5(b)(i)	- concentration decreases (1) - position of equilibrium shifts to left AND fewer moles of gas on left hand side (1)
5(b)(ii)	- concentration increases (1) - position of equilibrium shifts to the right AND forward reaction is endothermic (1)
5(b)(iii)	- no change (1) - catalysts do not affect position of equilibrium (1)
5(c)(i)	carbon dioxide
5(c)(ii)	- greenhouse gases absorb thermal energy (from the Earth) (1) - and one mark each for any two of: - energy (from the sun) absorbed by Earth's surface - Earth emits or reflects thermal energy (greenhouse gases) reduces or stops thermal energy loss (into space) - increasing in amount of greenhouse gas results in a higher atmospheric temperature

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 13

10 marks

2024 May/June Paper 41 Question 4

Main topic: Topic 6: Chemical reactions
Answer source label: 0620_s24_ms_41.pdf

4(a)	• The rate of forward reaction equals the rate of the reverse reaction(1) • concentrations of reactants and products are no longer changing(1)
4(b)(i)	• same number of gas moles on both sides of the equilibrium • OR • same number of gas molecules on both sides of the equilibrium
4(b)(ii)	• iodine particles or molecules (forced) closer together • OR • same number of iodine particles or molecules in a smaller volume
4(c)	• endothermic
4(d)	• -1 (1) • zero (1)
4(e)(i)	• copper
4(e)(ii)	• no effect
4(e)(iii)	• (activation energy is) lower

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 14

10 marks

2024 May/June Paper 43 Question 4

Main topic: Topic 6: Chemical reactions
Answer source label: 0620_s24_ms_43.pdf

4(a)	• (rates are) equal • (concentrations are) no longer changing
4(b)	• more molecules of nitrogen dioxide per unit volume • OR • same number of nitrogen dioxide molecules in a smaller volume
4(c)	• (the forward reaction is) endothermic
4(d)(i)	• increases • no • change • increases • increases
4(d)(ii)	• (the oxidation number of sulfur in S) = zero • (the oxidation number of sulfur in SO₃) = +6

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 15

19 marks

2024 Oct/Nov Paper 42 Question 3

Main topic: Topic 6: Chemical reactions
Cross-topic links: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w24_ms_42.pdf

3(a)	• attraction between molecules • attraction is weak
3(b)	• $2\text{S} + \text{Cl}_2 \rightarrow \text{S}_2\text{Cl}_2$
3(c)	• S with one dot-cross bonding pair with each Cl • four non-bonding dots for S • six non-bonding crosses for each Cl
3(d)	• decrease concentration of S_2Cl_2 / P and / or Cl_2 • increase temperature
3(e)(i)	• the minimum energy that colliding particles must have to react
3(e)(ii)	• E_a
3(e)(iii)	• increases no change • increases increases • increases
3(f)	• oxidation number of S goes from +1 • oxidation number of S goes to +2 • oxidation number of Cl (goes from -1) to 0 (in Cl_2) • oxidation numbers of Cl AND S increase

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 16

15 marks

2024 Oct/Nov Paper 43 Question 5

Main topic: Topic 6: Chemical reactions
Answer source label: 0620_w24_ms_43.pdf

5(a)(i)	oxygen gas escapes from the flask
5(a)(ii)	- the hydrogen peroxide is used up - OR - all the hydrogen peroxide has reacted or decomposed
5(a)(iii)	time intervals are too large
5(b)(i)	- kinetic energy of particles increases - frequency of collisions between particles increases - more / higher percentage OR higher proportion OR higher fraction of particles have energy greater than / equal to - activation energy - OR - more of the collisions / higher percentage OR higher fraction or higher proportion of collisions have energy greater than or - equal to activation energy
5(b)(ii)	- graph starts at same mass and has steeper gradient - levels off at the same mass but earlier
5(c)(i)	- oxidation number of manganese +4
5(c)(ii)	no change
5(d)(i)	not a closed system
5(d)(ii)	forward reaction is exothermic
5(d)(iii)	(to the) right - there are fewer gas molecules on the right

Easy explanation / reminder

- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.

END OF ANSWER

Question 17

14 marks

2025 Feb/March Paper 42 Question 4

Main topic: Topic 6: Chemical reactions

Cross-topic links: Topic 5: Chemical energetics, Topic 2: Atoms, elements and compounds

Answer source label: 0620_m25_ms_42.pdf

4(a)	• no reactants or products can enter or leave
4(b)	• enthalpy change (of reaction)
4(c)	• (The value of ΔH is) negative
4(d)	• +105 (kJ / mol)
4(e)	• decreases • increases • increases • no change
4(f)	• Bond energy in breaking bonds • = $1075 + 240 = (+) 1315$ • Bond energy of $\text{COCl}_2 = 1315 - (-105) = 1420$ • $\text{C}=\text{O} = 1420 - 2 \rightarrow \text{C}-\text{Cl}$ • = $1420 - (2 \rightarrow 340) = (+) 740$
4(g)	• both single bonding dot and cross pairs correct • double $\text{C}=\text{O}$ bond dot and cross pairs are correct • Complete diagram is correct • February/March 2025

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 18

17 marks

2025 Oct/Nov Paper 43 Question 4

Main topic: Topic 6: Chemical reactions
Cross-topic links: Topic 3: Stoichiometry, Topic 7: Acids, bases and salts
Answer source label: 0620_w25_ms_43.pdf

4(a)(i)	• rates are equal • concentrations are no longer changing
4(a)(ii)	• decreases • decreases • no change
4(a)(iii)	• forward reaction is exothermic
4(b)	• USUAL METHOD • 2490 • 2670 • -180 • ALTERNATIVE METHOD • 850 • 1030 • -180
4(c)	• 3 dot and cross bonding pairs • one lone pair on the N atom (to complete the N octet) • three lone pairs on each Cl atom (to complete each Cl octet)
4(d)(i)	• solid disappears • effervescence/bubbling/fizzing
4(d)(ii)	• $0.005 / 5 \rightarrow 10^{-3}$ (mol) • $0.0025 / 2.5 \rightarrow 10^{-3}$ (mol) • 60 (cm³)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER